TING-LUN SU

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EDUCATION

B.E. in Intelligent Automation Engineering Jun. 2022 – 2026

National Chin-Yi University of Technology (NCUT), Taichung, Taiwan

GPA: 3.36 / 4.00

Advisor: **Prof. Ta-Jen Peng**, Department of Intelligent Automation Engineering

RESEARCH INTERESTS

Focused on the deep integration of AI and industrial manufacturing — encompassing LLM-driven automation, optimization-based robotic path planning, and the development of scalable smart factory system integration.

PUBLICATIONS

Journal Articles

[1] T.-J. Peng, P.-J. Chen, T.-L. Su, C.-C. Lin, "Automated spray path planning based on Bayesian optimization and ant colony optimization," The International Journal of Advanced Manufacturing Technology, Springer, vol. 139, pp. 5491–5509, Aug. 2025. https://doi.org/10.1007/s00170-025-16162-x

[2] T.-J. Peng, P.-J. Chen, T.-L. Su, C.-C. Lin, "Optimization of automated forging release agent spray path based on ant colony algorithm," Journal of Automation Intelligence and Robotics, TAIROA, No. 51, pp. 114–120, Dec. 2024.

Conference Papers

[3] Z.-L. Sun, T.-S. Liaw, C.-L. Liu, T.-L. Su, C.-C. Wang, T.-J. Peng, "Green machining optimization and burr treatment in mill-turn operations," 27th Annual Meeting of the Chinese Society of Mechanism and Machine Theory, CSMMT Abstract No. 039, Nov. 2024.

[4] T.-L. Su, C.-H. Du, C.-H. Lai, "Development of a contactless AIoT smart data management hybrid vending system," 17th Intelligent Living Technology Conference, ILT Paper No. 3C009, Jun. 2023.

[5] T.-L. Su, Q.-L. Liu, Y.-H. Lin, C.-H. Lai, "The function analysis of green electric automatic weeder," Conference on Green Technology Engineering and Applications, GTEA Paper No. GEES012, p. 214, May 2023.

PATENTS

Invention Patent – Mushroom Stem Cutting Machine (No. I891383) May 2025

Taiwan

Automated device designed to precisely cut mushroom stems, improving efficiency and consistency in agricultural processing.

Utility Model Patent – Mushroom Stem Cutting Machine (No. M659734) Aug. 2024

Taiwan

Mechanical design enhancing stability and cutting performance of the mushroom stem processing system.

AWARDS & HONORS

First Place – 2nd Mitsubishi Electric Smart Automation Innovation Practice Competition (MECA) 2026
Judge's Special Award – MECA Cup Global Finals 2026 — Mitsubishi Electric (Japan) Global 2026
Second Place – Smart Manufacturing Talent Cultivation Program Machine Tool PBL Project Competition 2026
Second Place – National Vocational and Technical College Student Practical Project Competition 2025
First Place – 13th Intelligent Automation Equipment Invention Awards 2024
Third Place – Taiwan Society of Precision Engineering 113th Precision Engineering Project and Paper Award 2024
Honorable Mention – Mitsubishi Electric Contest of Automation (1st) 2024
Honorable Mention – Mitsubishi Electric CNC Smart App Creative Development Competition 2024
Science and Technology Humanities Award – 27th TDK Cup National Collegiate Creative Design Competition 2023
3 Gold, 2 Silver, 4 Honorable Mentions – National Project and Science Fair Competitions 2020 – 2022

RESEARCH & PROJECT EXPERIENCE

Visual Measurement Inspection Station & Generative AI Navigation System 2025 – Present

Industry-Academia Collaboration | TURVO International Co., Ltd. | Taichung, Taiwan

- Led project planning, technical design, and stakeholder communication with industrial partners.
- Developed core software integrating LLM and Model Context Protocol (MCP) for intelligent navigation assistance, achieving a 33.9% reduction in token cost via hybrid model design.
- Reduced single-part inspection time by 52.3% and workpiece defect rate by 22.7% through automated visual measurement integration. [System in Action]

Smart CNC Flexible Manufacturing Factory (Phase 1–3) 2024 – 2025

Industry-Academia Collaboration | IC Star Industrial Co., Ltd. | Taichung, Taiwan

- Executed a three-phase smart manufacturing project deploying 20 CNC machines and 10 robotic arms across 10 production lines, redesigning Plant A and building Plant B with full system integration.
- Led a 42-member cross-disciplinary team covering robotics, CNC communication, and control design, achieving an 80% reduction in required manpower. [Phase 1~2 System in Action]、[Phase 3 System in Action]

Automated Robot Arm Forging Spraying System with AI Path Generation (Phase 1–2) 2023 – 2024

Industry-Academia Collaboration | IC Star Industrial Co., Ltd. | Changhua, Taiwan

- Designed AI-based ant colony optimization for robotic forging spraying path planning, achieving 97.3% trajectory matching accuracy and 83% shorter line-switching setup time.
- Improved production efficiency by 27%, reduced forging defect rate to 65 PPM (91% reduction), and increased energy utilization by 21% across three automated forging lines.
- Showcased results at the 2024 Taipei International Automation Exhibition, initiating collaboration with a local forging factory for further joint development. [System in Action]

Development and Commercialization of Automatic Mushroom Stem-Cutting System 2022 – 2025

University Social Responsibility (USR) Project | Xinshe, Taiwan

- Successfully mass-produced the smart mushroom stem-cutting machine; adopted by 3 farms, reducing required farm labor by 36% and maintaining mushroom damage rate within ±1%.
- Secured 2 patents (invention + utility model) for the pressing and auto-sorting mechanisms, improving cutting consistency and agricultural processing efficiency.
- Presented the system at the 2025 Taipei USR EXPO, demonstrating sustainable agricultural automation and AI-based cutting control to national stakeholders. [System in Action]

RESEARCH & PROJECT EXPERIENCE

3D Vision-Guided Picking & Robotic System Integration Development 2022 – Present

Collaboration with LIEBHERR-Verzahntechnik GmbH & PONGI Gear Expert Co., Ltd. | Taichung, Taiwan

- Integrated and optimized Liebherr LHRobotics.Vision random picking system supporting bin depths up to ~1 m, with scan-and-plan cycle time of 0.6–7 s.
- Achieved ~99% robot compatibility and ≈98% collision avoidance success rate; provided client-side technical support to accelerate system adoption. [System in Action]

TECHNICAL SKILLS

Programming: Python, C#, C++, MATLAB

Automation & Robotics: PLC programming, Robot Arm control, CNC systems, Systems integration, CAD

AI / ML: AI visual detection , LLM integration, MCP systems, algorithm optimization

CERTIFICATIONS & LICENSES

- AI Application Planner – Licensed Technician, Ministry of Economic Affairs (iPAS)
- Photovoltaic System Installation – Licensed Technician Class B, Ministry of Labor (WDA)
- Industrial Wiring – Licensed Technician Class C, Ministry of Labor (WDA)
- Commercial Wiring – Licensed Technician Class C, Ministry of Labor (WDA)
- Master of Computer Certificate – Class C, Chinese Computer Education Association
- SAKURA SCIENCE Exchange Program Member, Japan Science and Technology Agency

LANGUAGES

Chinese: Native

English: Proficient – TOEIC 660 (actively improving)

German: Beginner (actively learning)

REFERENCES

Ta-Jen Peng, Ph.D. – **Thesis Advisor**

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